SPL1 Project Proposal Form, 2025
Institute of Information Technology (IIT)
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Project Description:

The project aims to explore and implement the **Lagrangian Relaxation algorithm** as a method to solve classical **NP-hard optimization problems** efficiently. NP-hard problems are computationally challenging, and finding exact solutions for large instances is often expensive in terms of time and computational resources. The Lagrangian Relaxation technique provides an approach to approximate optimal solutions by relaxing certain constraints and solving the resulting simpler subproblems iteratively.

In this project, I will apply the Lagrangian Relaxation algorithm to attempt to solve well-known NP-hard problems. The candidate problems include:

- **0–1 Knapsack Problem** – optimizing the selection of items to maximize profit under a weight constraint.
- **Capacitated Facility Location Problem (CFLP)** – determining optimal facility locations and allocations to minimize total cost under capacity constraints.

The objective is to:

- Formulate each selected problem mathematically.
- Apply **Lagrangian Relaxation** to decompose the problem into manageable subproblems.
- Compare the algorithm's performance and solution quality against standard exact or heuristic approaches.

This project will combine **theoretical understanding of combinatorial optimization** with **practical implementation using C/C++/Java and CPLEX** for solving the relaxed subproblems. The results will demonstrate how the Lagrangian Relaxation framework can provide high-quality approximate solutions to NP-hard problems in reasonable computation time.

Languages or Tools to be used:

- C/C++/Java
- IBM ILOG CPLEX Optimization Studio

Supervisor's Name: Shah Mostafa Khaled, Ph.D.

Signature of the supervisor:__________________________

Date: _________________________

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor:_______________________

Date: _________________________

Midterm Presentation Feedback: